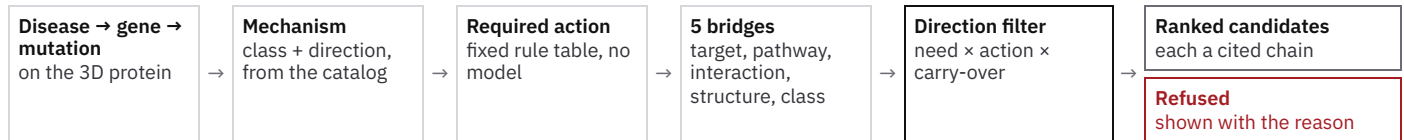




CHALLENGE TACKLED

About five thousand rare diseases come down to one broken gene, and fewer than one in twenty has a treatment. **Our user is a translational researcher with a gene and no therapy:** to find what might help one protein they must check every molecule recorded as acting on it, one database at a time, and judge whether each pushes the protein the right way or the wrong way. Helix covers the IUIS 2024 inborn-errors-of-immunity catalog: **511 genes, 604 diseases**.



Nine Omnigent agents — a supervisor and eight specialists — drive the same tool layer through a recorded loop.

TOOLS / ML MODELS USED

- **Claude Sonnet 5** — reasoning, every agent, both arms
- **Omnigent 0.16.0** — supervisor + 8 specialists
- **ChEMBL** — action type + activities: the direction signal
- **Reactome, IntAct, STRING** — pathway, interaction
- **Open Targets** — held-out ground truth
- **AlphaFold DB, ESMFold v1** — structures
- **Boltz-2** — binding affinity, pending a GPU backend
- **AlphaMissense, ProtVar** — variant effect
- **PrankWeb** — pockets; **Mol*** — 3D viewer
- **Europe PMC, OpenAlex** — literature
- **FastAPI** — API and discovery engine
- **Next.js** — web app

WHAT WORKED WELL

Direction of effect is a hard filter, not a ranking factor. A molecule that lowers a protein's activity helps a disease where that protein is overactive and harms one where it is already too weak, so anything pushing the wrong way is refused outright and shown with its reason. For X-linked agammaglobulinemia the engine refuses **34 molecules**, among them all 20 that ChEMBL records as lowering BTK — ibrutinib included, a cancer drug in use today.

The same rule still finds drugs: with the APDS link withheld, **leniolisib returns at rank 4 of 27** through a chain of three cited records, and for STAT1 gain of function baricitinib returns at **rank 2 of 23, aimed at JAK1 upstream**, not at STAT1. **All 4 stored controls pass.** One request settles **410 molecule-direction judgments** from 492 upstream records over 8 databases.

WHAT WAS CHALLENGING

Measuring the engine caught a real bug in our own filter. Run across all 604 diseases with the known drug links held out, it refused **plerixafor for WHIM syndrome** — the drug used for that disease. ChEMBL records its one CXCR4 mechanism as PARTIAL AGONIST, while the four activities ChEMBL holds for that same pair are all IC50: one curated field said raise, four measurements said lower, and the filter believed the field. The fix — **a rejection may no longer rest on an action type that another record of the same molecule and protein contradicts** — puts plerixafor back at rank 8 and makes the case a permanent fourth control.

Recall is low, and reported as it came out: 12 of 226 held-out pairs (**5.3%**) over 32 evaluable diseases, median rank 5.5, 1 false rejection. The cause table for the 213 misses shows most are antibodies, replacement proteins, transplant conditioning or drugs with no ChEMBL mechanism record at all — things a mechanism engine is right to stay silent about.

The ablation went against our own agent team, and we kept the result. Same tools, model and budget, 13 variants per arm: the lab took a median **198.5 s against 84.2 s** for one generalist agent, slower in 13 of 13 pairs. It bought breadth — 10.69 against 7.46 databases cited — and all 26 runs produced a complete cited record.

HOW WE SPENT THE TIME

 one day, 2026-10-03, from git log

12:45	Foundation — backend core, design system, shell	19:11–19:43	Interface — three stages, type scale, new mark
15:04	Platform — engines, journey pages, simple UI	20:00	Lab as the demo centrepiece
15:07–15:41	Benchmark: 26 runs, agent team against one agent	20:06	Measure the engine over 604 diseases
16:30	Lab, measured comparison, plain-language pass	20:23	Fix the filter , re-measure, classify misses
18:49	Discovery engine — bridges, direction filter	20:51	MIT relicence

If we had 24 more hours, we would add a functional-counterpart bridge: the 61-pair "no bridge path" bucket holds amiloride on ENaC, the channel that counterbalances CFTR — a relationship no physical-interaction record carries.

Every row is a hypothesis built from records and labelled as one. No treatment, no dose, no advice; nothing validated in a laboratory.